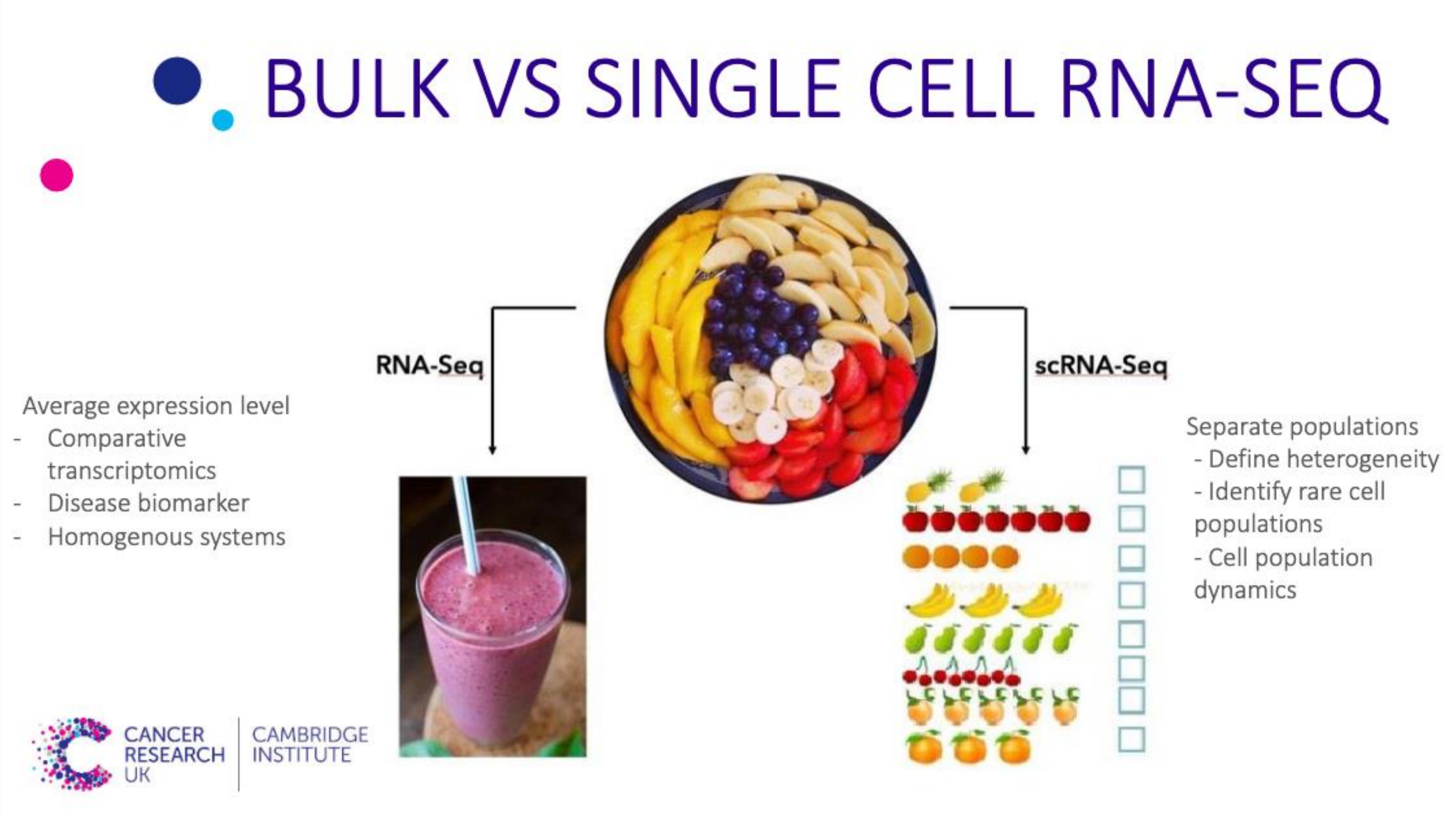
AI for (single-cell) transcriptomics workshop

Joseph Ng
Hana Andrew

Ctrl-Alt-Biology Workshop series

Supported by UCL Grassroots Research Culture Seed Fund

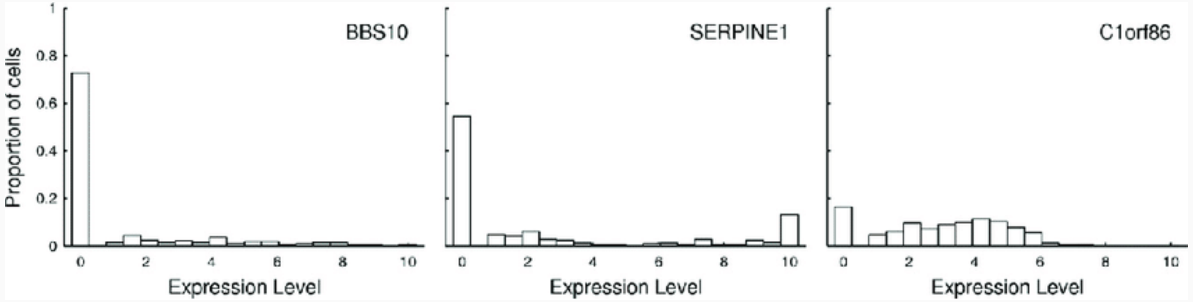
UCL

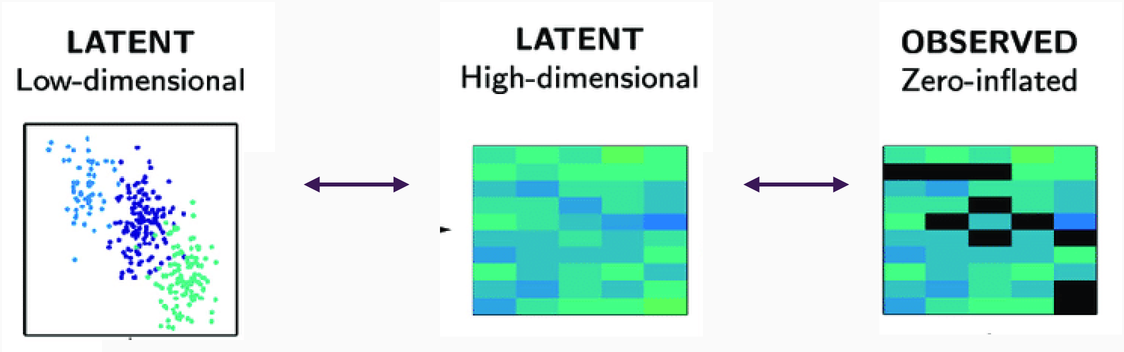


Taken from https://carpentries-incubator.github.io/scrna-seq-analysis/01-introduction/index.html

Challenges with single-cell transcriptomics

- Output: genes × cells count matrix (typically 20 000+ genes, thousands to millions of cells)
- Typical scRNA-seq matrices are 90–95% zeros — most genes appear "not expressed" per cell (**Sparsity**)
- Two sources: biological (gene truly off) vs technical dropout (low-capture mRNA molecules)
- Consequences: violates standard assumptions in methods like PCA; many statistical tests assume non-zero counts





Batch effects

- Technical variation from different labs, protocols, sequencing platforms, or capture dates
- **Significant problem for scRNA-seq data** which are already very noisy – Batch signal can dominate biological signal (ie cells from the same type but different batches cluster apart)
- Integration across datasets (multi-study, atlas-building) demands principled batch-aware models

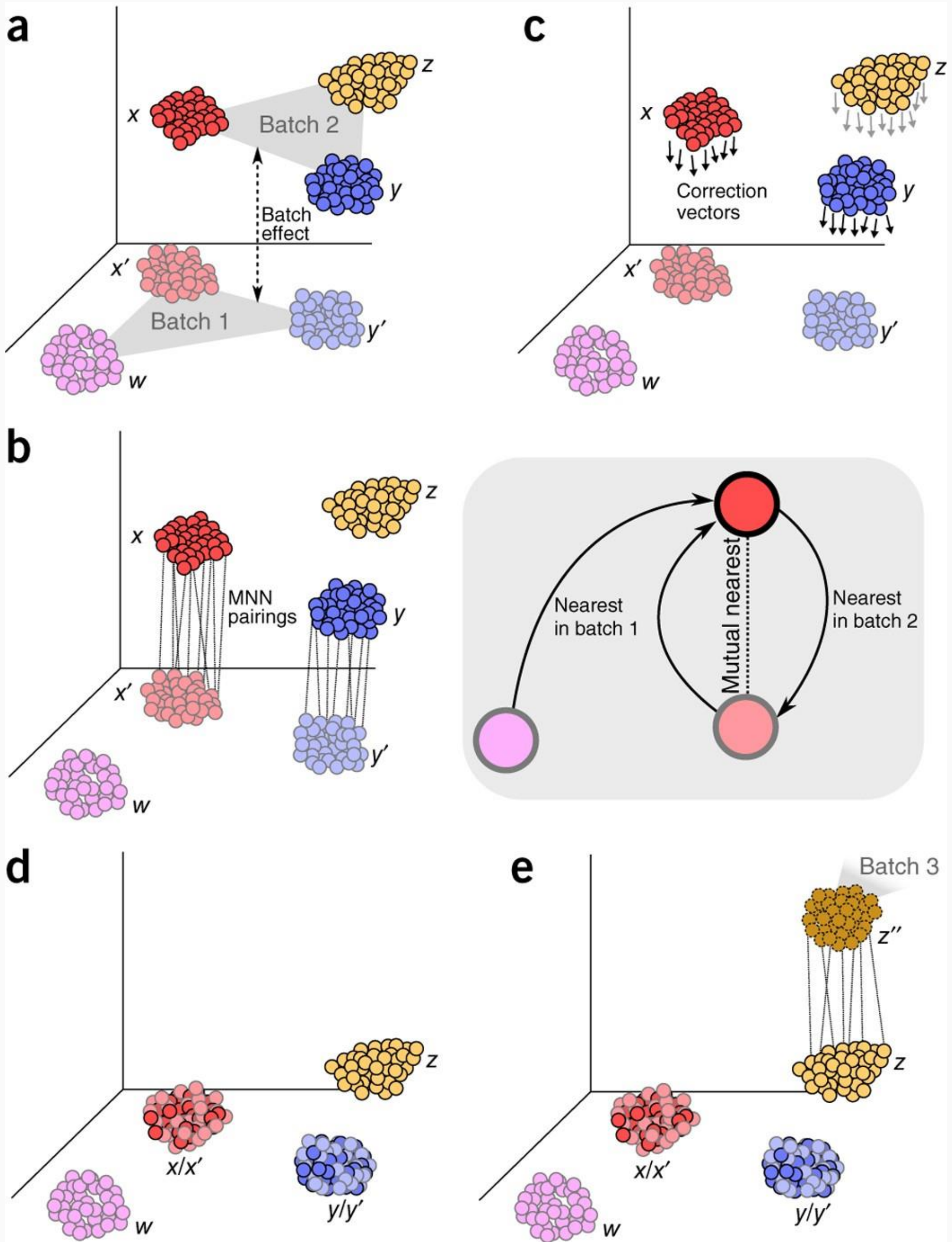
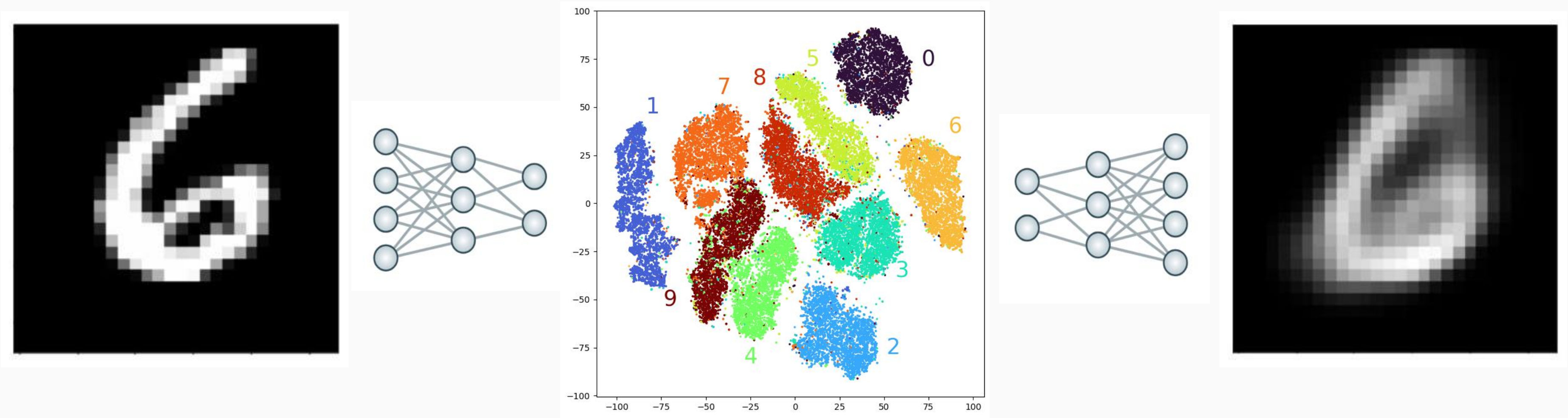


Figure from Haghverdi *et al. Nat Biotechnol* (2018). https://doi.org/10.1038/nbt.4091

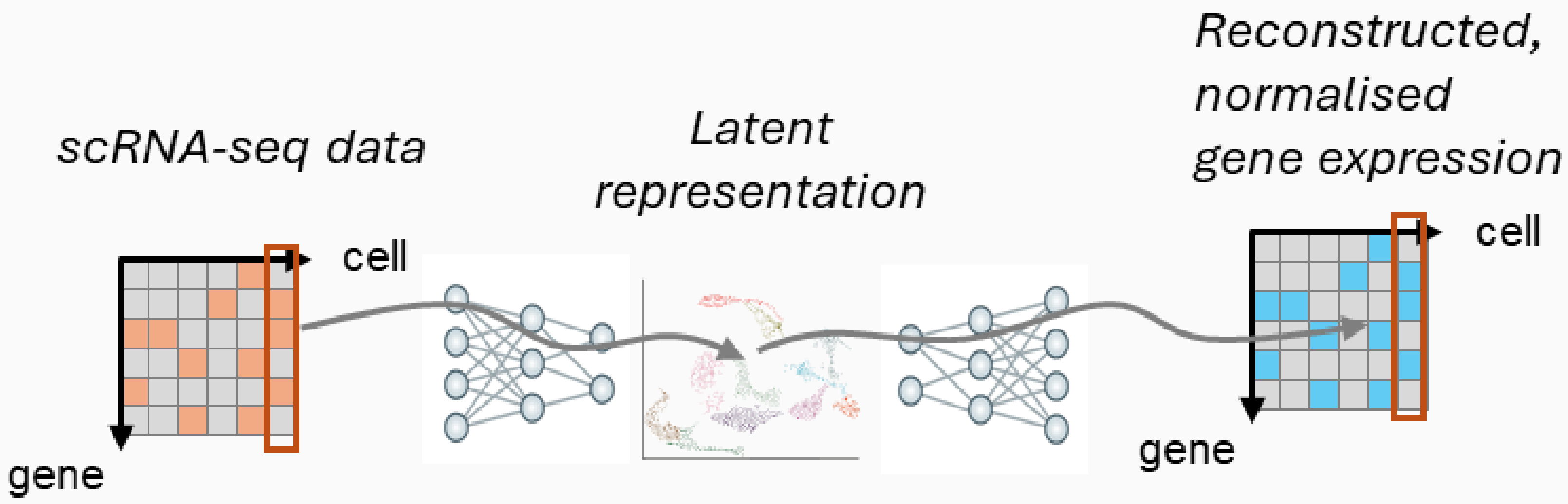
Autoencoders

- Encoder compresses high-dimensional input into a low-dimensional latent
- Trained end-to-end to minimise reconstruction loss — the **bottleneck** forces a compact representation
- “Generative”



Variational autoencoders (VAEs): state of the art for single-cell transcriptomics AI models

- "variational" AE (VAE) enabling generation and uncertainty
  - We have prior knowledge over how transcript counts (etc) should be distributed!
  - The model learns to reconstruct parameters of these distributions
- "Latent representation" in the middle captures differences between observations (cells) which can be used for downstream applications



Related models in the scVI family

scANVI — semi-supervised extension; incorporates cell-type labels where available, improves integration when label information exists

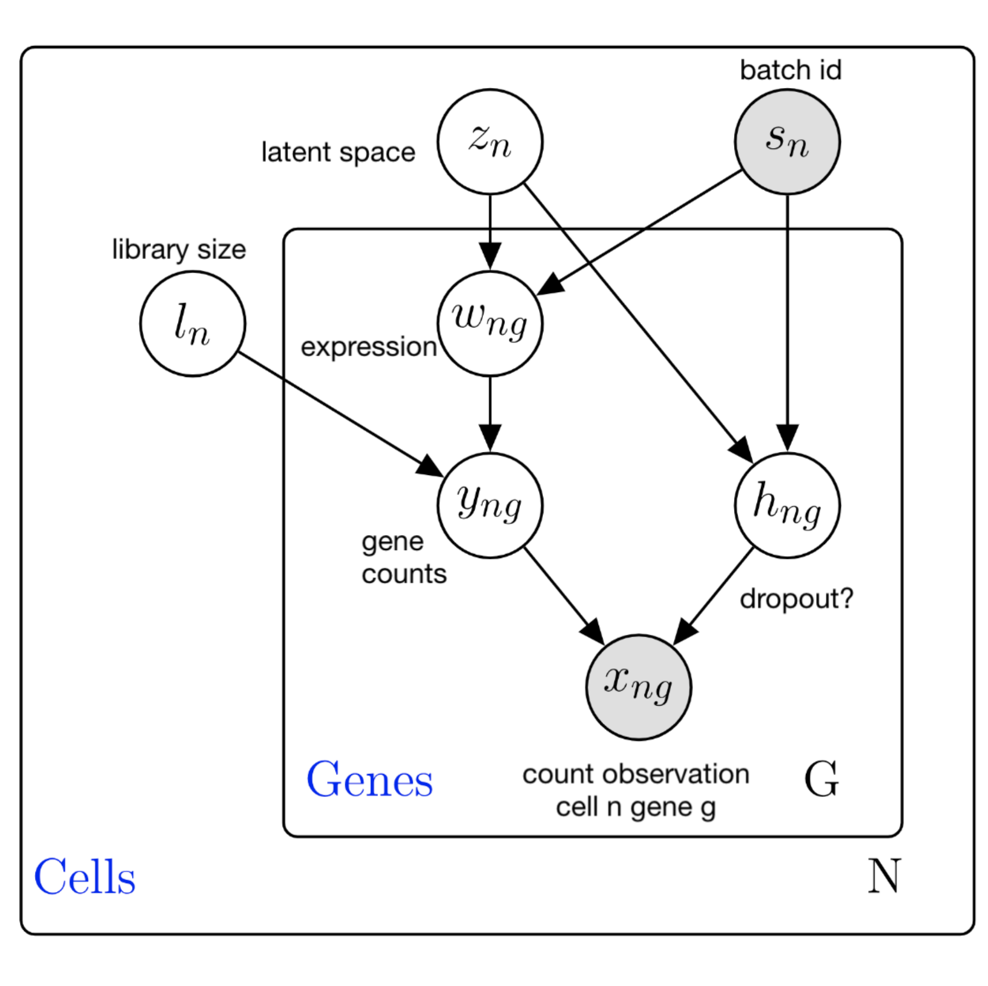
totalVI — joint model for RNA + protein (CITE-seq), with separate likelihoods per modality

peakVI — analogous model for chromatin accessibility (scATAC-seq) with a Bernoulli/Poisson decoder

MultiVI — multi-modal extension handling unpaired or partially paired multi-omic data

Batch correction & mapping new data

- Batches are explicitly modelled – the latent space should capture differences *while correcting for batches*
- Mapping new data: ***transfer learning***
  - *a* pre-trained reference model is extended with new batch-specific encoder weights for query data while freezing shared weights
  - **Label transfer**: query cells inherit cell-type annotations from the reference via KNN in latent space
  - Works with scVI, scANVI, totalVI, and other VAE-based models — not architecture-specific



Today

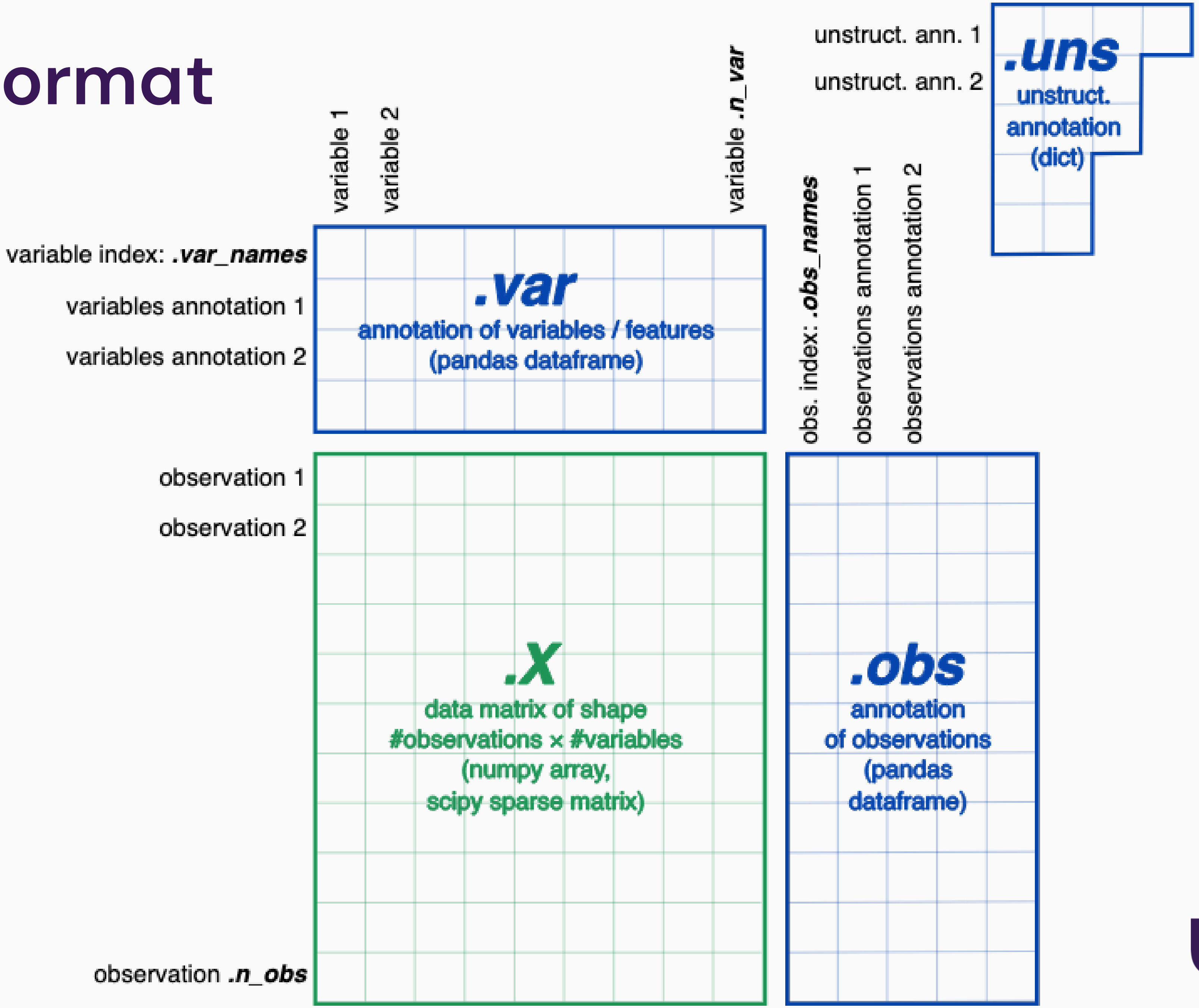
Our workbook contains a standard example of what you might want to do with applying scVI-type models on typical scRNA-seq tasks:

- Dimensionality reduction
- Training a model for reference dataset → Data integration with query dataset & label transfer.

We will look at scenarios where this works well and those where there may be problems!

- Defining 'marker' genes via differential gene expression
- Gene annotation (incl. some LLM based approaches)

AnnData format



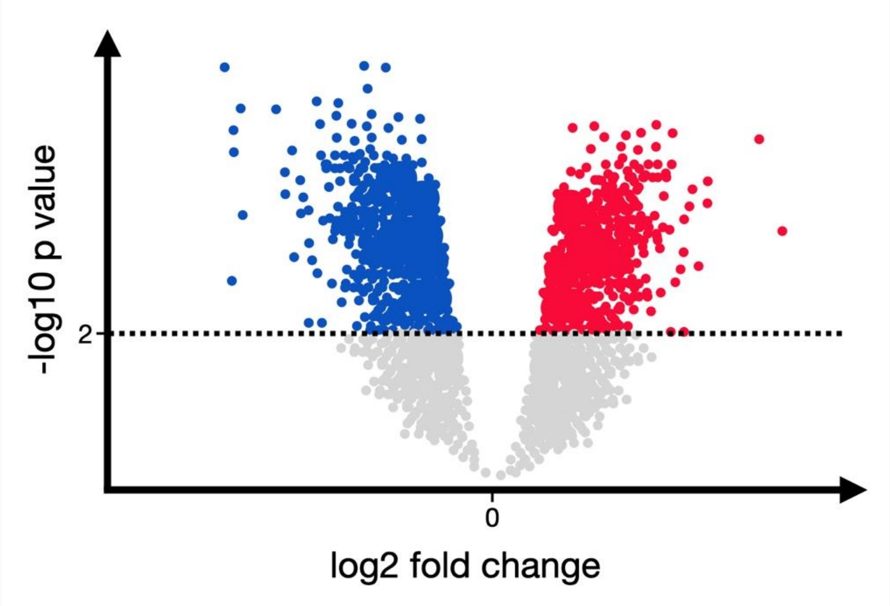
Differential gene expression

Wilcoxon rank sum is a standard method (default in Scanpy) to identify genes differentially expressed in two populations

Asks: If I randomly pick a Basal 1 cell and a Basal 2 cell, how often is expression higher in Basal 1?

We get tiny p values, this method treats each cell as an independent observation

This can lead to inflated significance as cells originating from the same sample/donor/batch aren't independent



Wilcoxon top 10 up in Basal 1

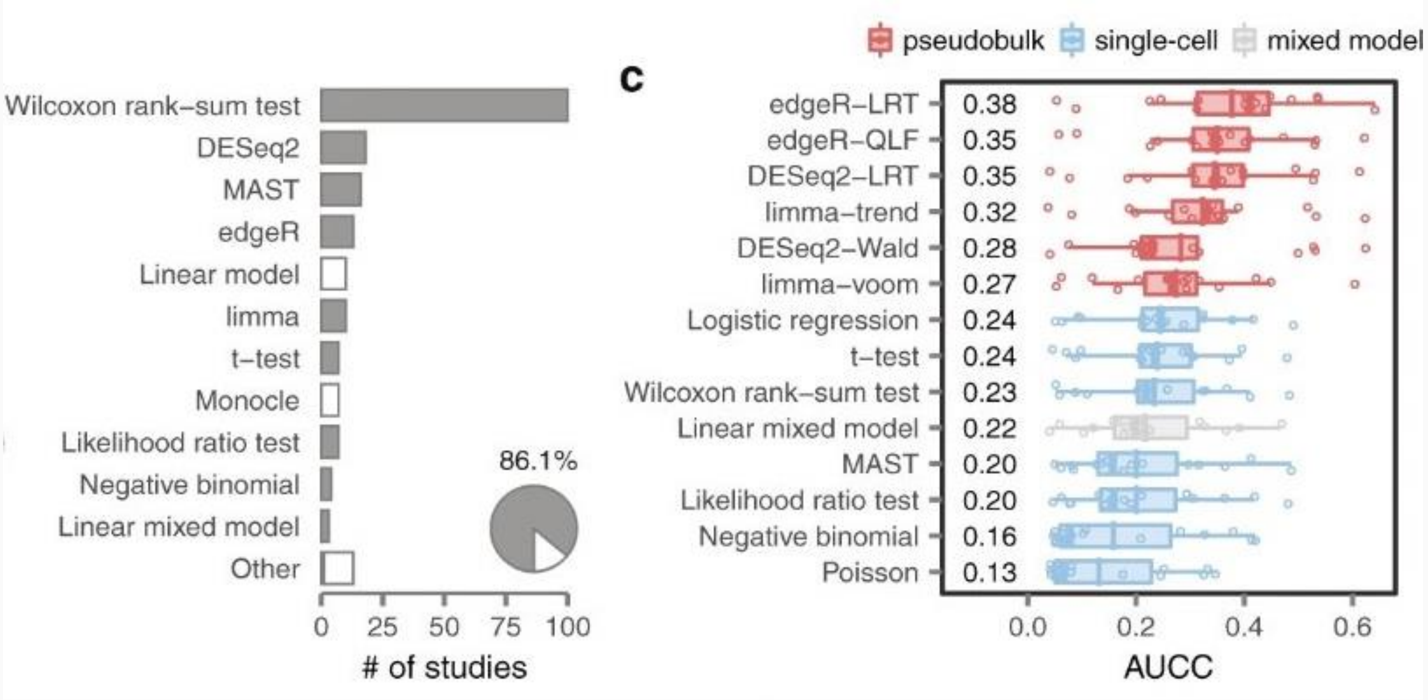
	names	logfoldchanges	pvals_adj
0	KRT15	1.053229	4.282494e-298
1	ZFP36L2	1.036440	2.782169e-273
2	DST	1.105281	6.897287e-257
3	JUNB	0.602197	4.608555e-232
4	CTGF	1.488313	1.586046e-198
5	F3	0.580268	4.281318e-189
6	ALDH3A2	1.080989	1.311411e-187
7	FOS	0.577772	7.972445e-185
8	CYR61	1.060034	8.105482e-179
9	BCAM	0.760209	1.907963e-172

Pseudobulk differential gene expression

DESeq2 balances differential effect size and batch variance

Asks: which genes are robustly different between Basal 1 and Basal 2 across batches?

A gene can only rank highly if fold change is in the same direction for all donors




```
Pseudobulk: 12 samples × 1025 genes
per group:
{'Basal_1': 6, 'Basal_2': 6}
Pseudobulk DESeq2 top 10 up in Basal 1
         log2FoldChange        pvalue          padj
index
GPR155         1.669845  1.527005e-34  1.203985e-32
KRT15          1.083035  1.271204e-31  9.307028e-30
BCAM           0.792804  9.624023e-27  4.932312e-25
DST            1.014668  1.210020e-25  5.637593e-24
EFEMP1         0.966124  9.126603e-24  3.897820e-22
CH25H          1.332076  3.997215e-22  1.470954e-20
PTRF           1.149762  1.260382e-21  4.454798e-20
POSTN          2.340980  1.738485e-20  5.241021e-19
DKK3           1.671304  2.726373e-19  7.552790e-18
COL14A1        2.952499  1.222972e-18  3.226739e-17
```

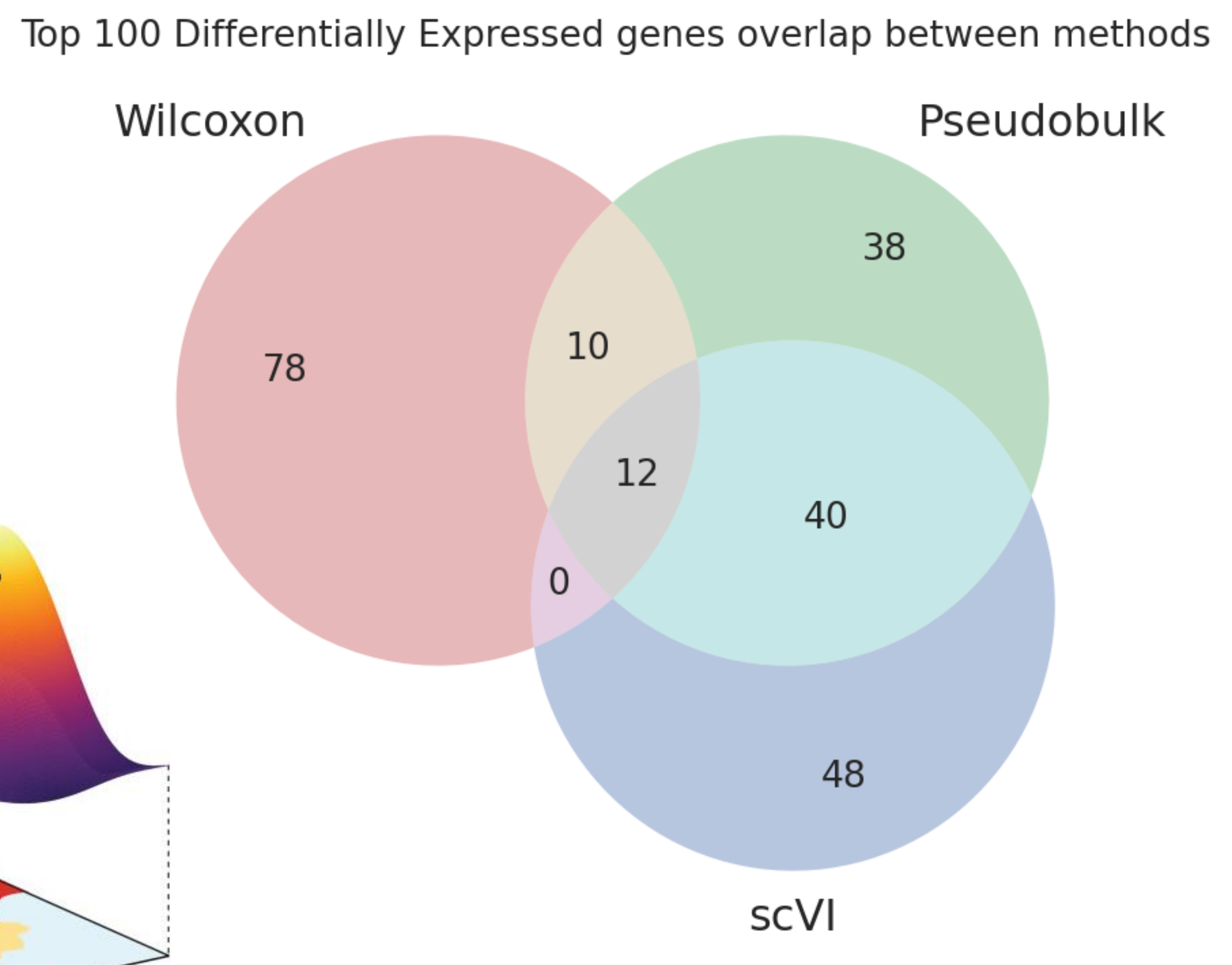
Squair, J.W *et al.* Confronting false discoveries in single-cell differential expression. *Nat Commun* (2021).

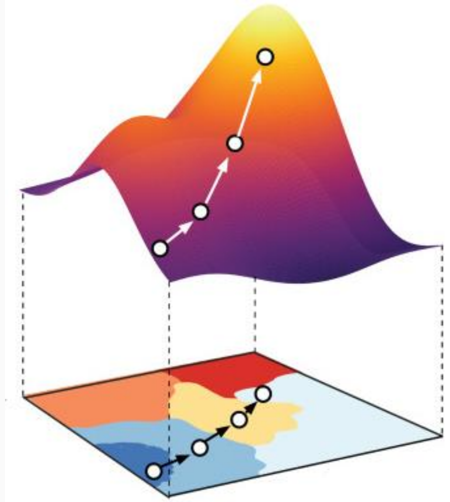
scVI differential gene expression

Instead of directly comparing noisy observed gene counts, scVI compares the inferred expression distributions from the latent model

DESeq2 assumes linear batch effects, scVI instead learns using neural networks that allow for joint modelling of multiple variation factors

scVI gives a probability that change exceeds a biologically meaningful threshold



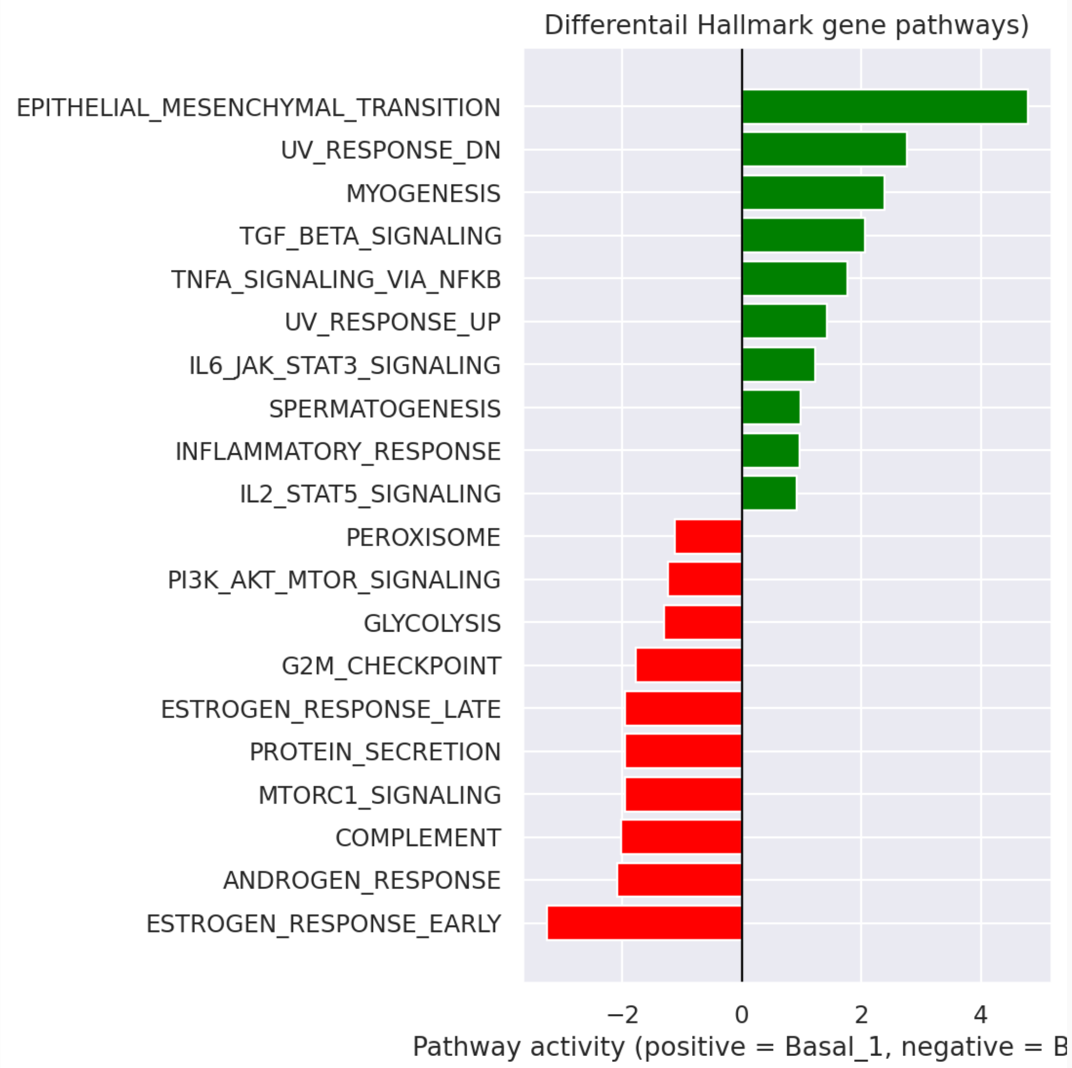


Interpretation of gene lists

We can ask whether genes that differentiate Basal 1 and 2 are part of gene pathways of curated known biological states/processes

decoupleR fits a linear model of gene differential expression stats to genes in the Hallmark gene resource (50 pathways)

The t statistic of the linear model is used to give an "activity score"

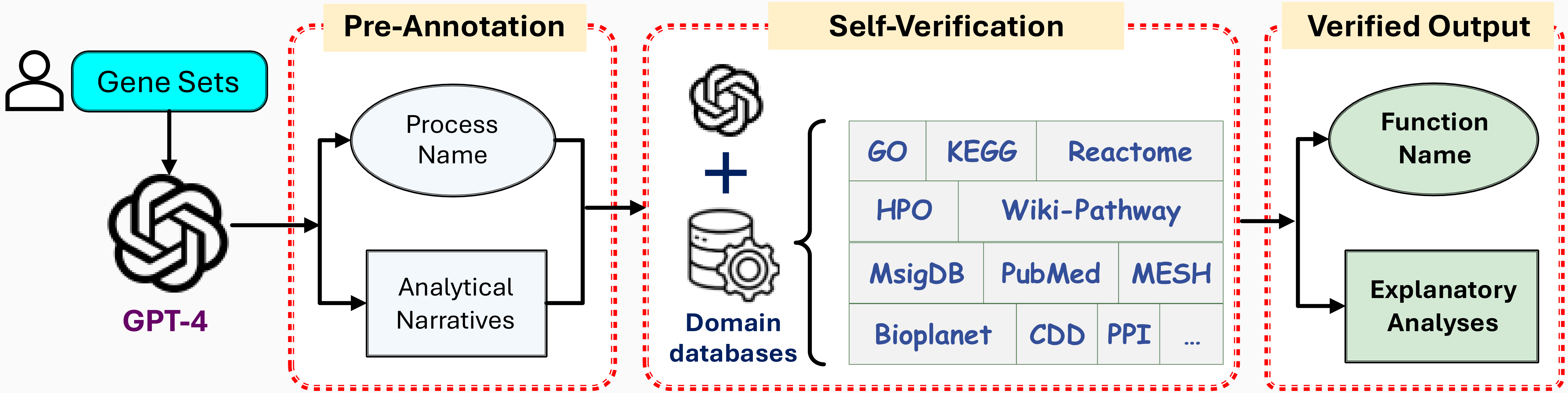


Interpretation of gene lists using LLMs

GeneAgent uses GPT-4 to return functions/processes for a list of genes

It has 4 stages of self verification to prevent hallucinations where claims made by the LLM's first output are extracted and queried against domain specific databases to support or refute the statement

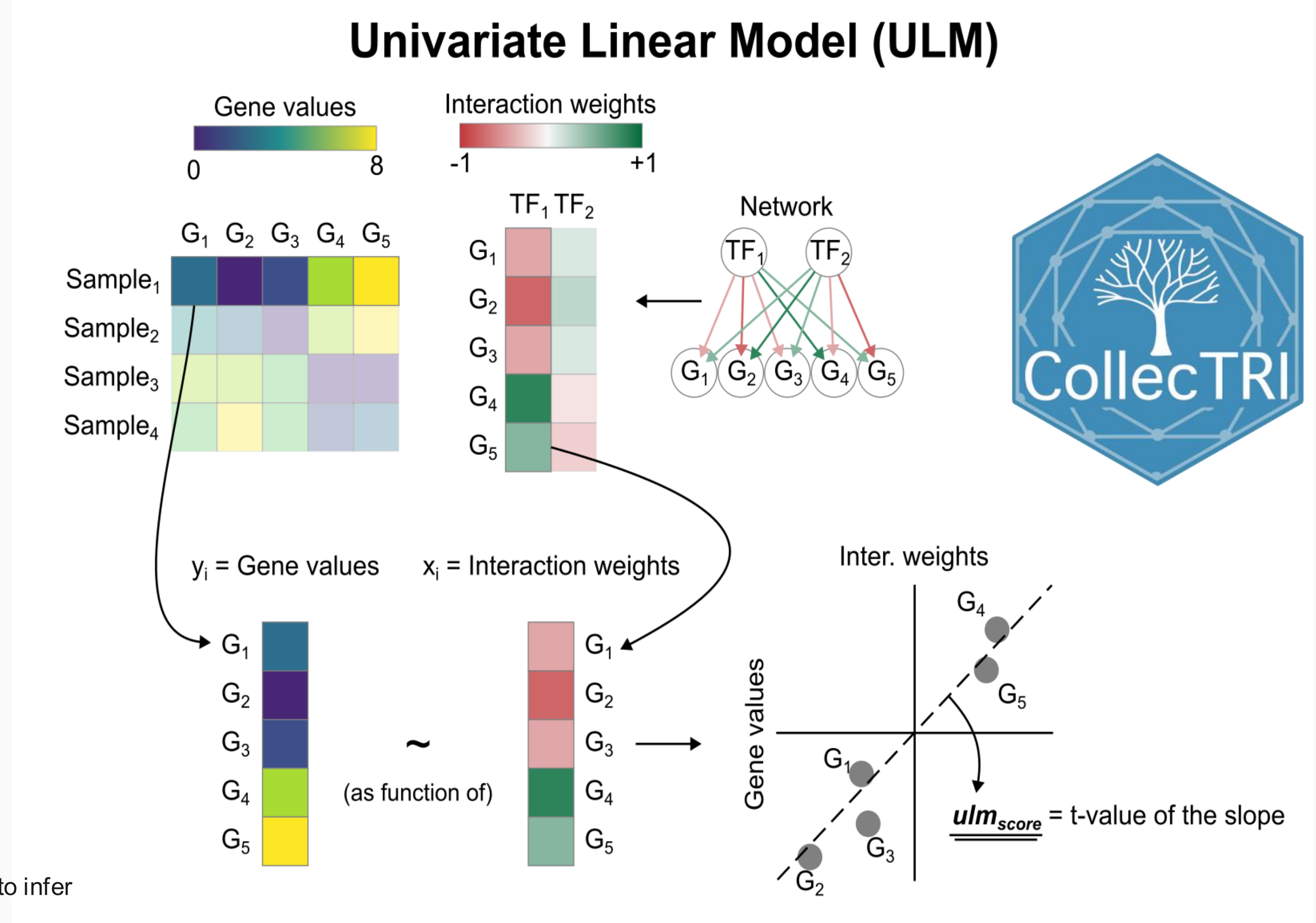
https://www.ncbi.nlm.nih.gov/CBBresearch/Lu/Demo/GeneAgent/



Transcription factor inference

Going beyond gene sets, we can ask whether genes known to be activated by a TF collectively more highly expressed than expected

Standard methods like decoupleR use similar methodology as gene set analysis but use known gene regulatory networks with interaction weights between TFs and genes to infer TF activity



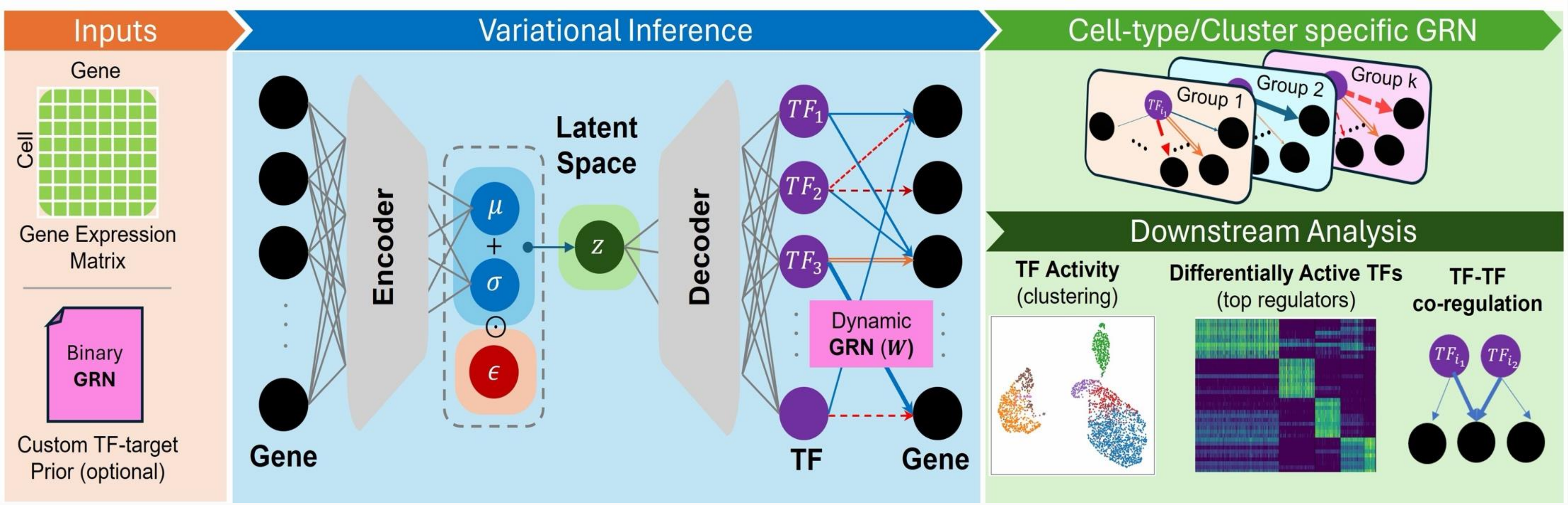
Badia-I-Mompel P, et al. decoupleR: ensemble of computational methods to infer biological activities from omics data. Bioinform Adv. 2022

scRegulate

Uses a VAE to model TF activities together rather than a linear model for each TF separately

The decoder connects TFs to genes using weights initialised from prior knowledge (collecTRI same as decouplR!)

But in scRegulate the GRN is not fixed, it can be trained and updated to better fit the gene expression profile



Mehrdad Zandigohar, Jalees Rehman, Yang Dai, scRegulate: single-cell regulatory-embedded variational inference of transcription factor activity from gene expression, *Bioinformatics* 2025

Survey

https://forms.cloud.microsoft/e/eR8E5DY8iR

We value your opinion – will help greatly in shaping future sessions!

Ctrl-Alt-Biology AI workshops - Survey (Transcriptomics)